MIL-C-13449A(MU)
17 August 1966
SUPERSEDING
MIL-C-13449(ORD)
21 May 1954

MILITARY SPECIFICATION

CARTRIDGE, CALIBER .30, CARBINE, TRACER, M27

1. SCOPE

1.1 This specification covers Cartridge, Caliber .30 Carbine, Tracer, M27 for use in the Caliber .30 Carbine.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition Through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

DRAWINGS

U. S. Army Munitions Command

D7640939	- Cartridge, Caliber .30, Carbine, Tracer M27

FSC 1305

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2. (Cont'd)

DRAWINGS (Cont'd)

U.S. Army Munitions Command

B8594160	- Cartridge, Caliber .30, Carbine, Tracer, M27, Steel Case
C7643674	- Classification of Cartridge Case Defects
D7553724	- Packing and Marking, Cartridge, Carbine, Caliber .30 in 10 Round Clips; in Bandoleers; in Box Ammunition, M2A1; in Wirebound Box
D7553902	- Packing and Marking, Cartridge, Carbine, Caliber .30; Cartons; Box, Ammunition, M2A1; Box, Wirebound
IEL-7640939	- Inspection Equipment, List for Cartridges, Caliber .30, Carbine, Tracer, M27
IEL-8594160	- Inspection Equipment List for Cartridge, Caliber .30, Carbine, Tracer, M27 (Steel Case)

PUBLICATIONS

U. S. Army Munitions Command
TECP 700-700 Vol. III - Manual of Test Methods for Small Arms Ammunition

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge shall comply with Drawing D7640939, or B8594160 as specified by the contract, referenced specifications and the following:

3.2 First article sample.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.3 Bullet extraction.- The force required to extract the bullet from the cartridge case shall be not less than 45 pounds.

3.4 Residual stress.- (brass case cartridges) The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.5 Accuracy.- The average of the mean radii of all targets of the sample cartridges, fired at 100 yards, shall not exceed 3.5 inches.

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3.6 Velocity.- The average velocity of the sample cartridges, conditioned at 68° to 72° Fahrenheit (F), shall be 1800 feet per second (ft/sec) plus or minus 35 ft/sec at 53 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 36 ft/sec.

3.7 Waterproof.- The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 5 pounds per square inch (psi) for 15 seconds.

3.8 Chamber pressure.- The average chamber pressure of the sample cartridges, conditioned at 68° to 72° F, shall not exceed 40,000 psi.

3.9 Trace.- When viewed at night from a line parallel to the plane of the trajectory at each point of observation, the bullet of the tracer cartridge shall exhibit a visible trace of full luminosity from a point not greater than 100 yards from the muzzle of the weapon to a point not less than 400 yards from the muzzle. The trace shall be of a level below full luminosity, and preferably invisible, from the muzzle of the weapon to a point at least 15 yards from the muzzle.

3.10 Salt spray (steel case cartridges).- No steel case shall show any corrosion products after the cartridge has been subjected to a 5 percent salt spray for 5 hours.

3.11 Function and casualty.- The cartridge shall function without casualty.

3.12 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.12.1 Metal defects.- The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.12.2 Foreign matter.- The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 1600 cartridges. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- The ammunition lot size shall be as agreed upon between the supplier and the contracting officer, provided the formation of the lot is in accordance with MIL-STD-105.

4.3.1.3 Lot identification.- Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the supplier's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

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4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The acceptable quality level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent. All non-conforming cartridges shall be rejected.

4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table I.

TABLE I

NO.	DEFECT AND METHOD OF INSPECTION Visual/	CRITICAL	MAJOR	MINOR	MAJOR or MINOR
	Cartridge				
1	Discolored, dirty, oily, smeared			X	
2	Corroded or stained, if etched		X		
	Case				
4	Round head		X		
5	Dent				X
6	Split case				
	in K, L or M location	X			
	in I or J location		X		
7	Perforated case	X			
8	Draw scratch				X
9	Scratch			X	
10	Beveled underside of head		X		
11	Scaly metal				X
12	No chamfer on head (rim)		X		
13	Fold			X	
14	Wrinkle			X	
15	Bulge			X	
16	Illegible or missing head stamp			X	
17	Defective head			X	
18	Defective mouth			X	
18A	Defective protective finish			X	

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4.3.2.1 Table I (Cont'd)

NO.	DEFECT AND METHOD OF INSPECTION	CRITICAL	MAJOR	MINOR	MAJOR or MINOR
	Bullet				
19	Dent			X	
20	Scratch			X	
21	Split bullet jacket		X		
22	Loose bullet		X		
23	Scaly metal				X
24	Upset (crooked) point			X	
25	Exposed steel (clad jac			X	
26	Flat point			X	
	Primer				
27	No primer	X			
28	Cocked primer	X			
29	Inverted primer	X			
30	Loose primer		X		
31	Nicked or dented primer			X	
32	No waterproofing material (primer pocket joint)			X	
	Gaging				
33	Total length		X		
34	Cartridge profile failure (requiring more than 20 lbs dead weight to insert in profile and alignment gage)		X		
35	Diameter of extractor groove, max.		X		
36	Diameter of extractor groove, min.			X	
37	Diameter of head		X		
38	Thickness of head		X		
39	Length of case		X		
40	Depth of primer		X		
	Weighing				
41	Weight, min. 2/		X		

1/ Refer to MIL-STD-636 (Caliber .30, Carbine) for visual defect standards for defects 1 through 32.

2/ Each lightweight cartridge shall be disassembled and the propellant weighed. Each such cartridge found to contain less than 7 grains of propellant shall be classed as a critical defect.

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4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

Table II

Test	Number of Cartridges	Requirement Paragraph
Bullet extraction 1/	25	3.3
Residual stress (mercurous nitrate) 1/	50	3.4
Accuracy 2/	90	3.5
Velocity 2/	20	3.6
Waterproof 3/	20	3.7
Chamber pressure 2/	20	3.8
Trace 4/	100	3.9
Salt spray (steel only) 2/	20	3.10
Function and casualty 5/ Carbine, Caliber .30, M2	600	3.11

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

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4.3.3.1 Table II (Cont'd)

3/ Failure of nine or more cartridges to comply with the applicable requirements shall be cause for rejection of the lot. If more than four, but less than nine cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if, in the combined first and second sample, nine or more cartridges fail to comply with the applicable requirement.

4/ The lot shall be rejected if more than twenty (20) shots fail to comply with the applicable requirements. No second sample permitted.

5/ The lot shall be rejected when function and casualty defects plus firing defects observed in all other firing tests exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the first tests exceeds the number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample consisting of double the quantities specified under function and casualty test shall be fired in the service weapon specified therefor. This procedure shall apply regardless of the weapon or weapons in which the firing defects occurred in the first test. If the total number of defects in th combined first and second sample exceeds the acceptance number fo. the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

4.3.3.2 Firing defects.- Firing defects and acceptance numbers shall be as specified in Table III.

TABLE III

Defects	Acceptance First Sample	Acceptance Cumulative (1st & 2nd Sample)
1. Misfire	1	2
2. Bullet remaining in bore 1/	0	-
3. Primer Leak		
a. Perforation in firing pin indent in primer cup	8	15
b. Escape of gas through primer cup other than 3a	4	7
c. Escape of gas around primer cup more than 50% of periphery	17	35
d. Blown primer or primer falls out of pocket on retraction of bolt	0	1
e. Primer remains in pocket, but is physically loose	4	7

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4.3.3.2 Table III (Cont'd)

Defects	Acceptance First Sample	Acceptance Cumulative (1st & 2nd Sample)
4. Case casualties		
a. Longitudinal split 2/		
(1) Mouth (I)	17	35
(2) Body (J)	9	17
(3) Body (K)	9	17
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 2/		
(1) Partial (J)	4	7
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
5. Failure to extract	0	1
6. Weapons stoppage 3/	0	1

1/ No second sample permitted. Lot shall be rejected.

2/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

3/ All stoppages attributable to the ammunition with the exception of misfire, complete rupture or failure to extract, observed in all tests shall be included.

4.3.4 Packaging, packing and marking inspection.- During or immediately prior to the packaging operation, 100 per cent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test, dummy grenade or cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than those listed shall be classed as a major defect. All non-conforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

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4.3.5 Inspection equipment.- The examination and tests shall be made using the equipment listed on IEL-7640939 or IEL-8594160 as applicable except as specified in 4.3.

4.4 Test methods and procedures.

4.4.1 Bullet extraction.- The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three nor more than six inches per minute.

4.4.2 Residual stress (mercurous nitrate).- Tests shall be conducted in accordance with TECP-700-700, Volume 3.

4.4.3 Waterproof.- The ammunition to be tested, not to exceed five cartridges at one time, shall be placed horizontally on a perforated tray. The tray shall be placed in a desiccator containing sufficient quantity of freshly boiled water to maintain a head of 2 to 2½ inches. The desiccator shall be evacuated to a pressure of 5 pounds per square inch (10 inches of mercury) below atmospheric pressure and held at that pressure for fifteen seconds. The number of bubbles liberated from the mouth or primer of each cartridge shall be observed.

4.4.4 Accuracy.- The test shall be conducted in accordance with Publication TECP 700-700 Vol. III.

4.4.5 Velocity.- The test shall be conducted in accordance with Publication TECP 700-700 Vol. III.

4.4.6 Chamber pressure.- The test shall be conducted in accordance with Publication TECP 700-700 Vol. III.

4.4.7 Salt spray.- Steel case cartridges shall be subjected to the salt spray test conducted in accordance with the Federal Test Method Standard No. 151, in a 5 percent solution, with the cartridges suspended bullet down for 5 hours.

4.4.8 Trace.- Observation for trace performance shall be conducted on a range of at least 400 yards length. All trace testing shall be performed at night. The cartridges shall be fired in a gun assembled with a barrel which has not previously been fired more than 2000 times. Observation for trace performance shall be made at the weapon and at points 15 yards, 100 yards and 400 yards beyond the muzzle of the gun on a line parallel to and approximately 50 yards from the plane of the trajectory. Lights or other suitable visible markers shall be placed opposite the 15, 100 and 400 yard observing points to insure a line of sight from each point perpendicular to the plane of the trajectory. The ammunition shall not be conditioned to temperature but shall be fired under existing conditions. Three cartridges shall be fired to sight, warm and foul the weapon prior to the test. The sample cartridges shall be fired in regular sequence permitting sufficient time between shots for each observer to record trace characteristics at each observation point.

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4.4.8 (Cont'd)

Each cartridge shall be considered as a defective only once in the event it fails at more than one observation point.

Observations for trace performance shall be as follows:

a. Observation at weapon position.- The observer at this point shall observe for any abnormal behavior as an informational check only. Record shall be made of erratic flight, bursting bullets, muzzle flash of obviously greater intensity than that produced by other bullets in the sample, or such other behavior as may be considered abnormal by the observer.

b. Observation at 15 yard position.- The observer, with line of sight established perpendicular to the trajectory by the light or visible marker, shall note the trace performance of the bullet from each sample cartridge. Trace of low level luminosity or absence of trace extending for at least 15 yards from the muzzle shall be considered acceptable. The cartridge shall be considered defective if the trace reaches full luminosity during the first 15 yards of the trajectory.

c. Observation at 100 yard position.- The observer, with line of sight established perpendicular to the trajectory by the light or visible marker, shall note the trace performance of the bullet from each sample cartridge. A trace which has attained full luminosity prior to 100 yards range shall be considered acceptable. The cartridge shall be considered defective when the bullet exhibits a low level luminosity trace or no trace.

d. Observation at the 400 yard position.- The observer, with line of sight established perpendicular to the trajectory by the light or visible marker, shall note the trace performance of the bullet from each sample cartridge. A trace of full luminosity extending to 400 yards range or beyond shall be considered acceptable. The cartridge shall be considered defective if the bullet exhibits a trace of low level luminosity or no trace.

e. "Full luminosity" is defined as the maximum level of sustained luminosity.

f. "Low level of luminosity" is a level obviously considerably lower in intensity than full luminosity.

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4.4.9 Function and casualty.- The test shall be conducted in accordance with Publication TECP 700-700 Vol. III. Four carbines shall be used. One hundred and fifty rounds shall be fired in each weapon. One-half of the number of cartridges for each weapon shall be fired full automatic action, and one-half fired semi-automatic action. 15 or 30 round magazines may be used.

4.4.10 Defect penalty.- In any ballistic test, except function and casualty, in which the occurrence of a firing defect listed in Table III prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A.- (worldwide shipment).- The cartridges shall be packed in accordance with Drawing D7553724 or D7553902.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the applicable drawing cited in 5.1.

6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Requirements for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

NOTE: Asterisks not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - MU

User
Air Force 70

Preparing activity:

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